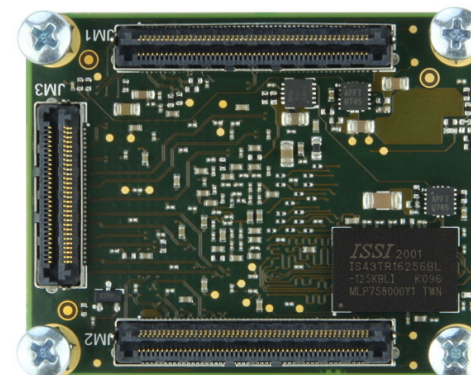
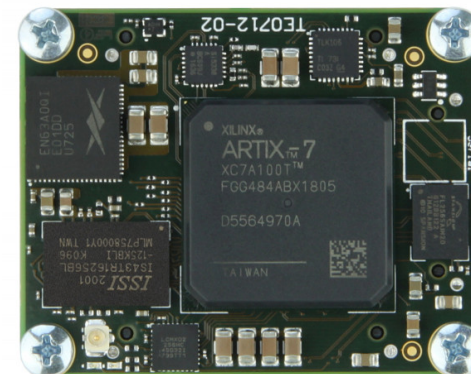
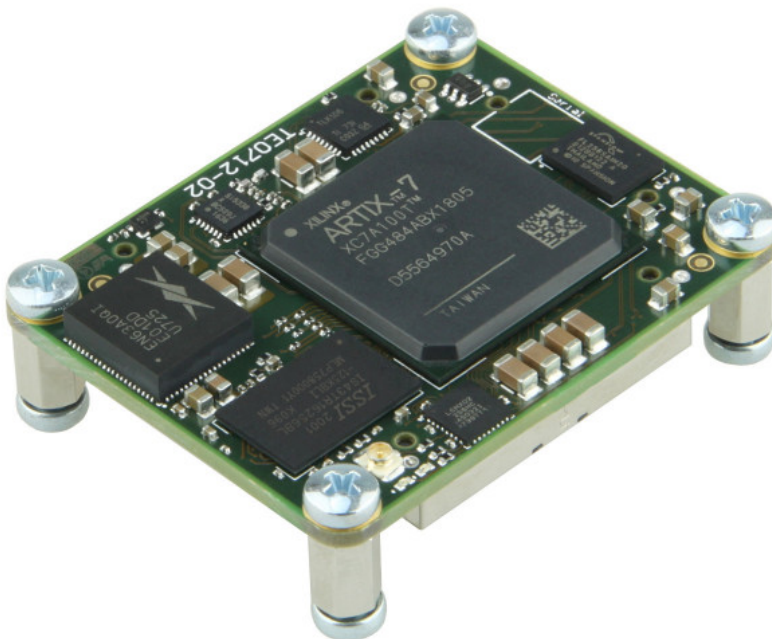




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Project is protected under copyright and we strongly and strictly prohibit the reverse engineering or recreation, even if the design is just adapted or modified. TE0712 is protected under such right and in case of plagiarism we will have to do anything necessary in order to protect our assets.

Schematics and other handouts serve for informational purposes only!

	Title: Legal Notices Modules		
	A4	Number: TE0712 81136-A	Rev. 03
	Date: 29.07.2026	Copyright: Trenz Electronic GmbH	Page 1 of 20
	Filename: Legal Notices Modules.SchDoc		

1		2		3		4		
A								A
B								B
C								C
D								D

REV	Description	
-01	Initial revision	
-02		
-03	1. Added Legal notices, project overview and revision changes. Updated page count and page order. 2. Added a D3 diode between the INIT and PROG_B signals to keep the FPGA in the reset state while PROG_B is low during the initial power-up. 3. Resistors R2 , R68 replaced by 2K2 (were 4K87) to improve I2C stability at higher baud rates. 4. Change obsolete ferrite beads BKP0603HS121-T to MPZ0603S121HT000. 5. Revised power supply circuit. Change obsolete components: - EN63A0QI - MP8869SGL-Z (U14); - EP53F8QI - MPM3834CGPA (U6 , U16). 6. Change Q1 power switch TPS27082LDDCR to MP5077GG-Z. 7. Added power monitors U10 , U11 STM6710LWB6F. System controller pin U3.25 connected to net PG_ALL instead of 3.3V . 8. U14 I2C interface connected to bus PLL_SDA / PLL_SCL U1B . Added table with device addresses on the I2C bus. A new device will be detected during a bus scan 9. Change capacitors in net "VIN" from 47 uF 6.3 V to 22 uF 10 V for C70 , C80 , C126 , C127 , C132 , C176 , C177 .	VY



Title:Revision History

A4

Number:TE071281I36-A

Rev.03

Date:29.07.2026

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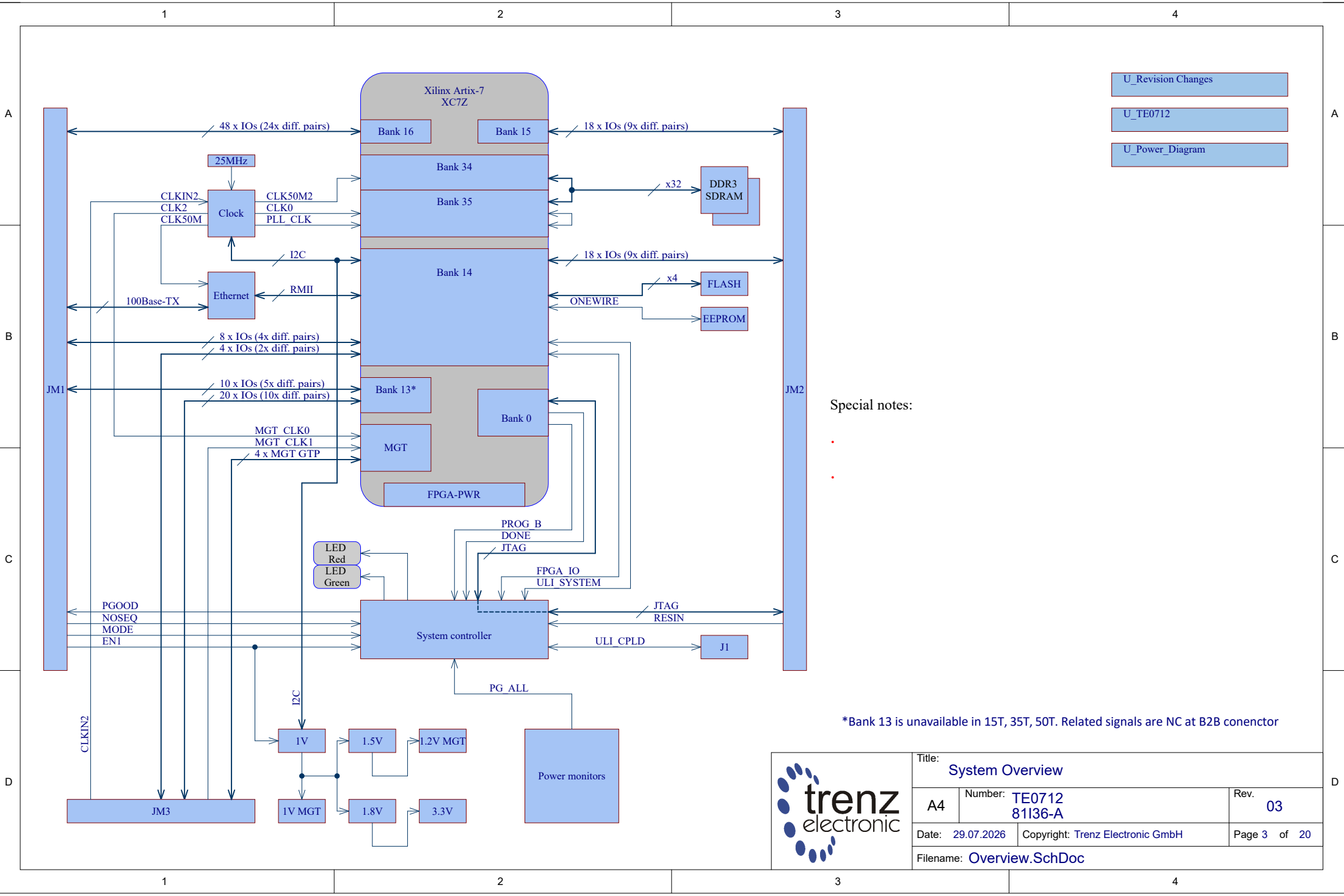
Page 2 of 20

Filename: Revision Changes.SchDoc

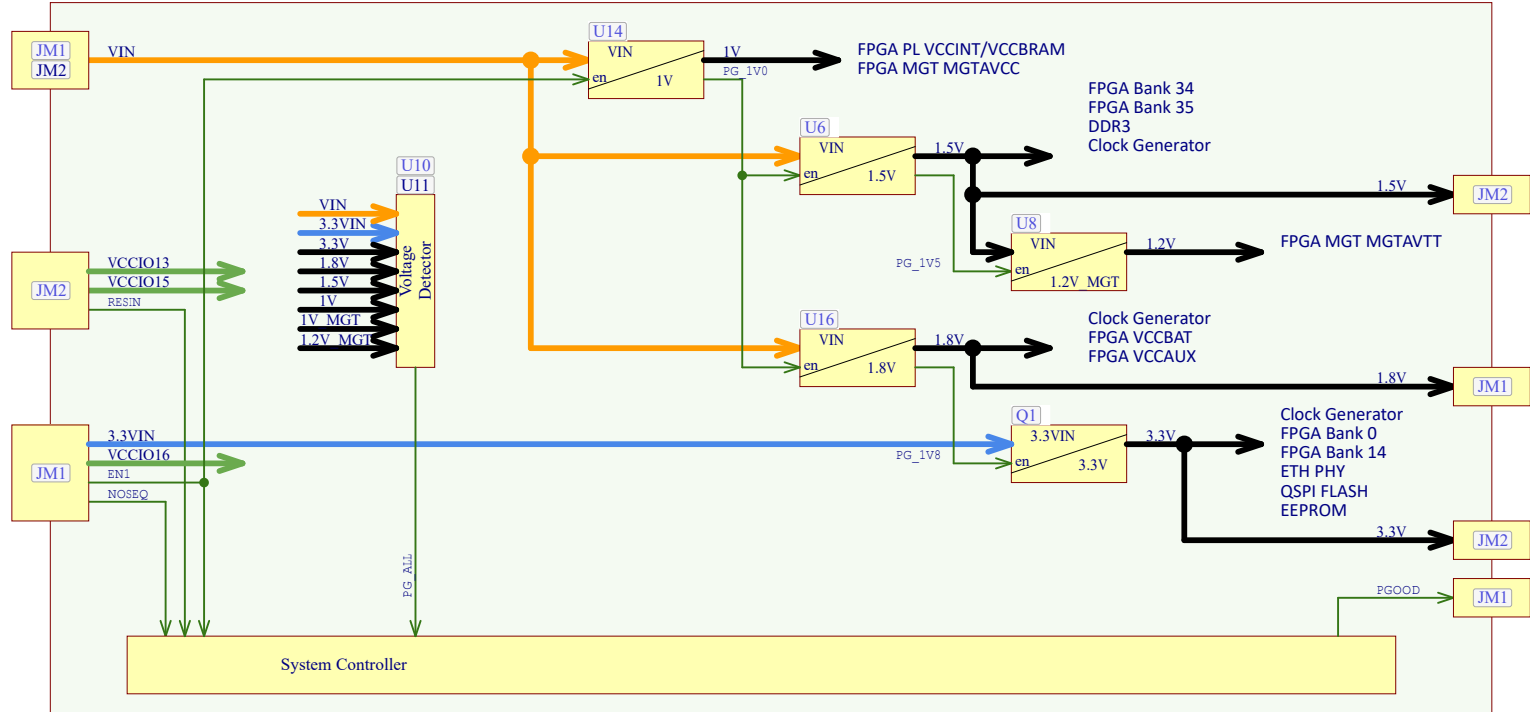
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REV	Description	
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	Title: Revision History		
	A4	Number: TE0712 81136-A	Rev. 03
	Date: 29.07.2026	Copyright: Trenz Electronic GmbH	Page 2 of 20
	Filename: Revision Changes.SchDoc		

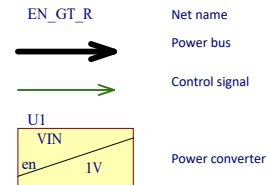


Power-on sequencing:



Recommended Operating Conditions

Power Rail	Direction	Range	Tolerance	Description	Note
VIN	IN	3.3 - 5V	+/-5%	Micromodule Power	Mandatory
3.3VIN	IN	3.3V	+/-3%	Micromodule Power	Mandatory
VCCIO13	IN	1.2 - 3.3V	+/-3%	HR IO Bank13	-
VCCIO14	IN	3.3V	+/-3%	HR IO Bank14	Fixed
VCCIO15	IN	1.2 - 3.3V	+/-3%	HR IO Bank15	-
VCCIO16	IN	1.2 - 3.3V	+/-3%	HR IO Bank16	-
1.5V	OUT	1.5V	+/-3%	For Carrier card Periphery	-
1.8V	OUT	1.8V	+/-3%	For Carrier card Periphery	-
3.3V	OUT	3.3V	+/-3%	For Carrier card Periphery	-
VREF_JTAG	OUT	3.3V	+/-3%	For Carrier card Periphery	Connected to 3.3V



	Title: Power Diagram	
	A4	Number: TE0712 81136-A
Date: 29.07.2026	Copyright: Trenz Electronic GmbH	Rev. 03
Filename: Power_Diagram.SchDoc		Page 4 of 20

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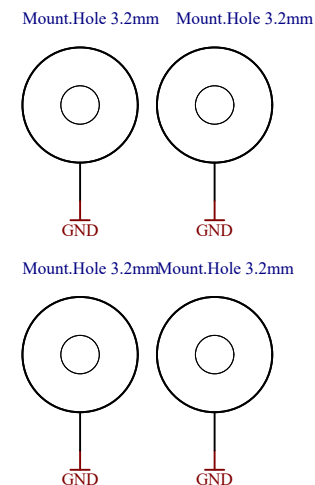
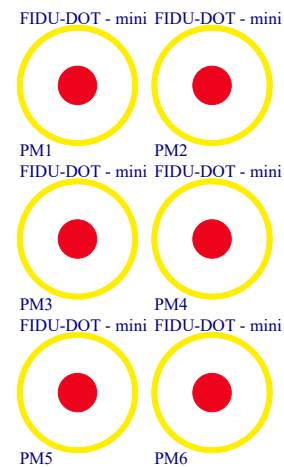
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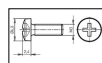
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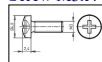
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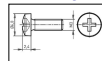
Top of Board



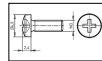
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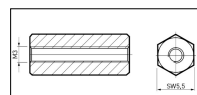
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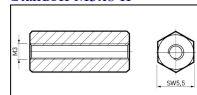
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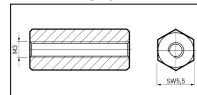
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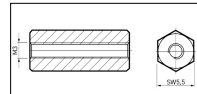
Standoff M3x8 II



Standoff M3x8 II



Standoff M3x8 II



Standoff M3x8 II



Screw M3x6



Screw M3x6



Screw M3x6



Screw M3x6

Serial
Serial
Serialnumber 6,3 x 6.3mm

SerialI
TE Address Overlay
LOGO ADDRESS



Title: TE0712		
A4	Number: TE0712 81136-A	Rev. 03
Date: 29.07.2026	Copyright: Trenz Electronic GmbH	Page 5 of 20
Filename: TE0712.SchDoc		

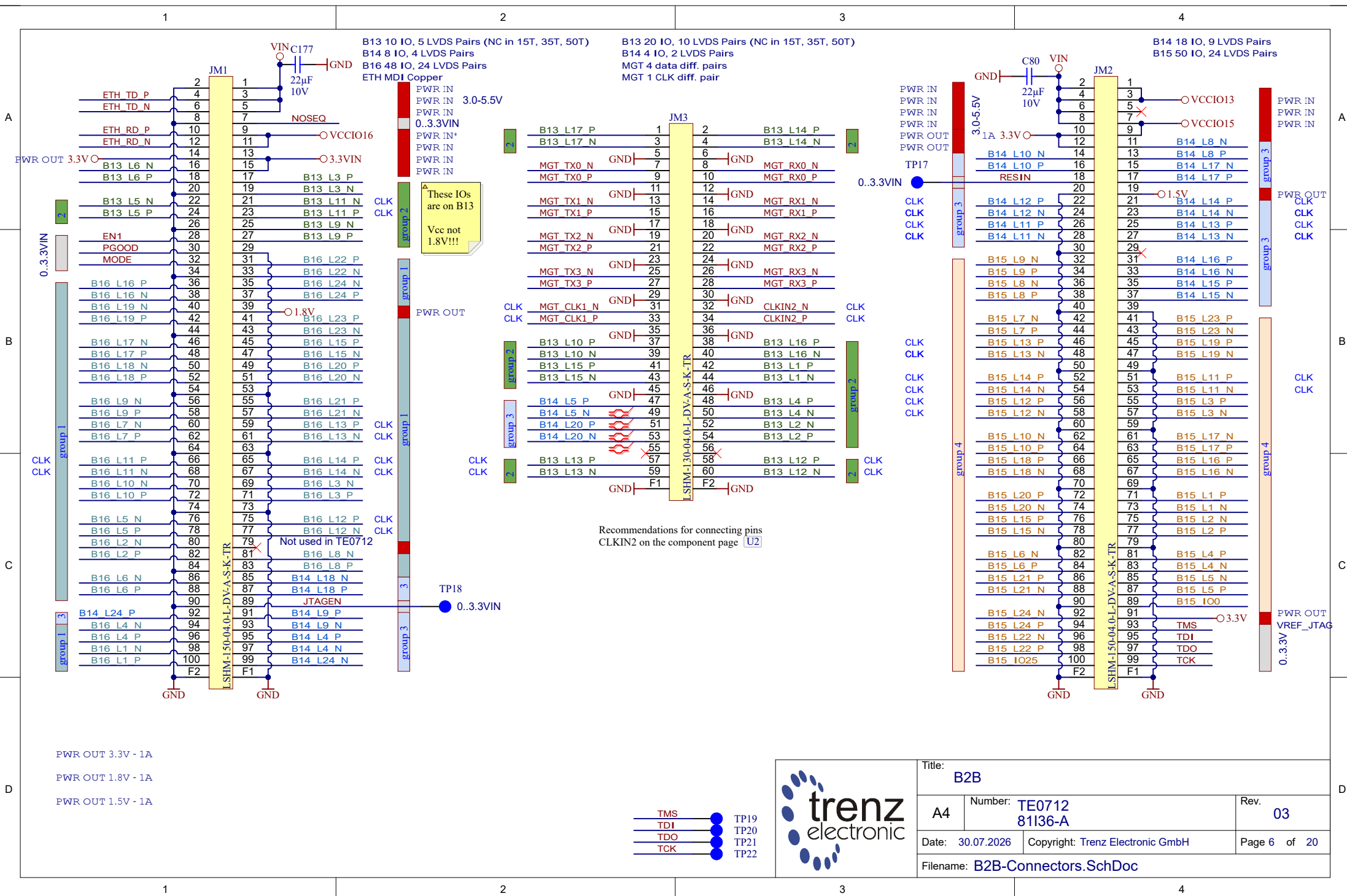
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Modified by	MR
Modified at	2021-02-16
SVN Revision	14002 [Locally Modified]

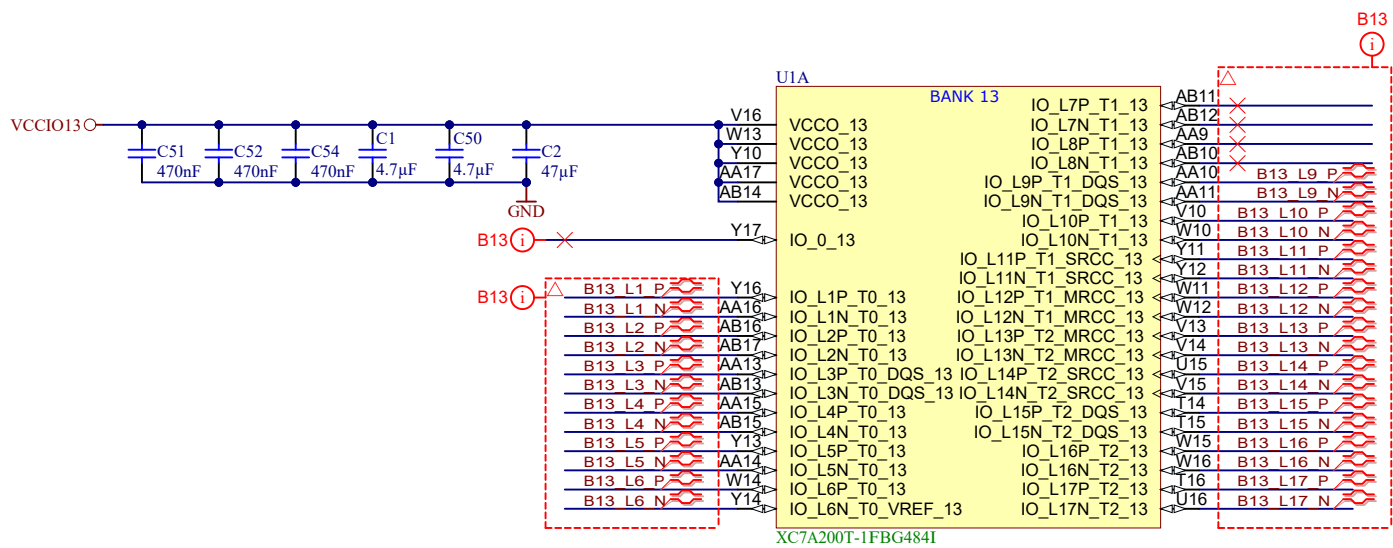
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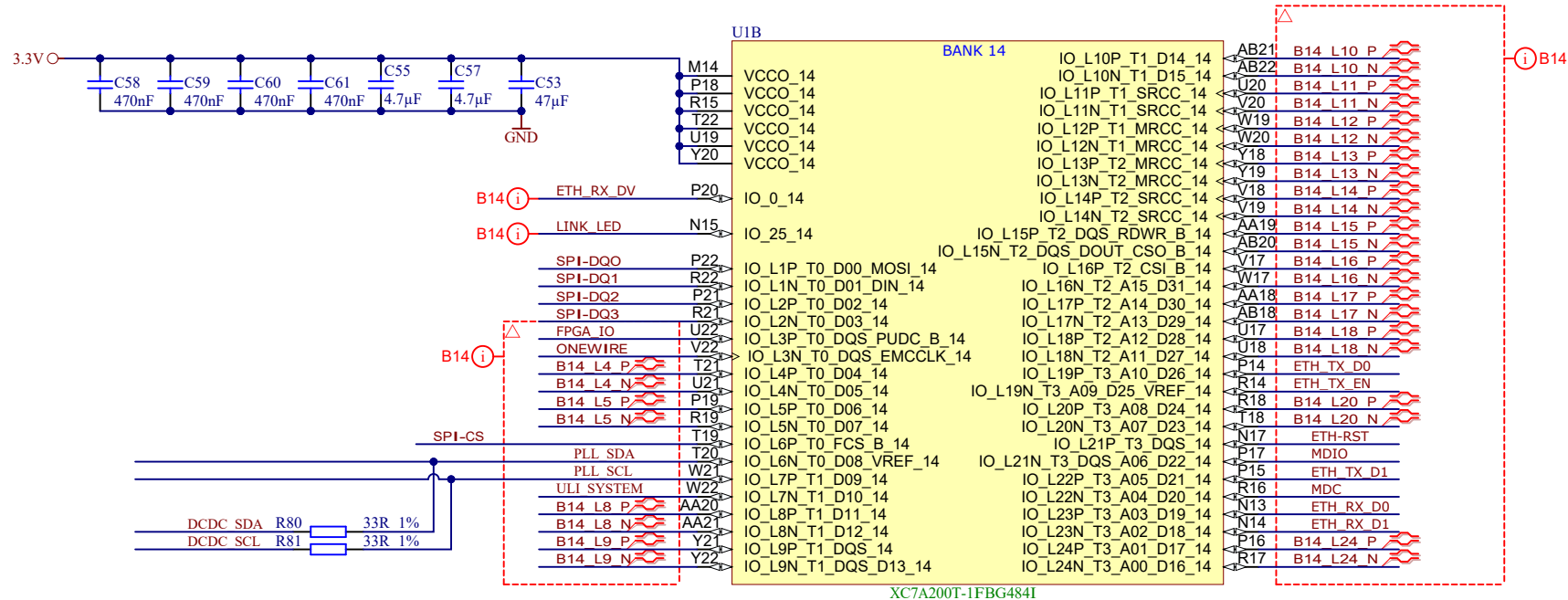
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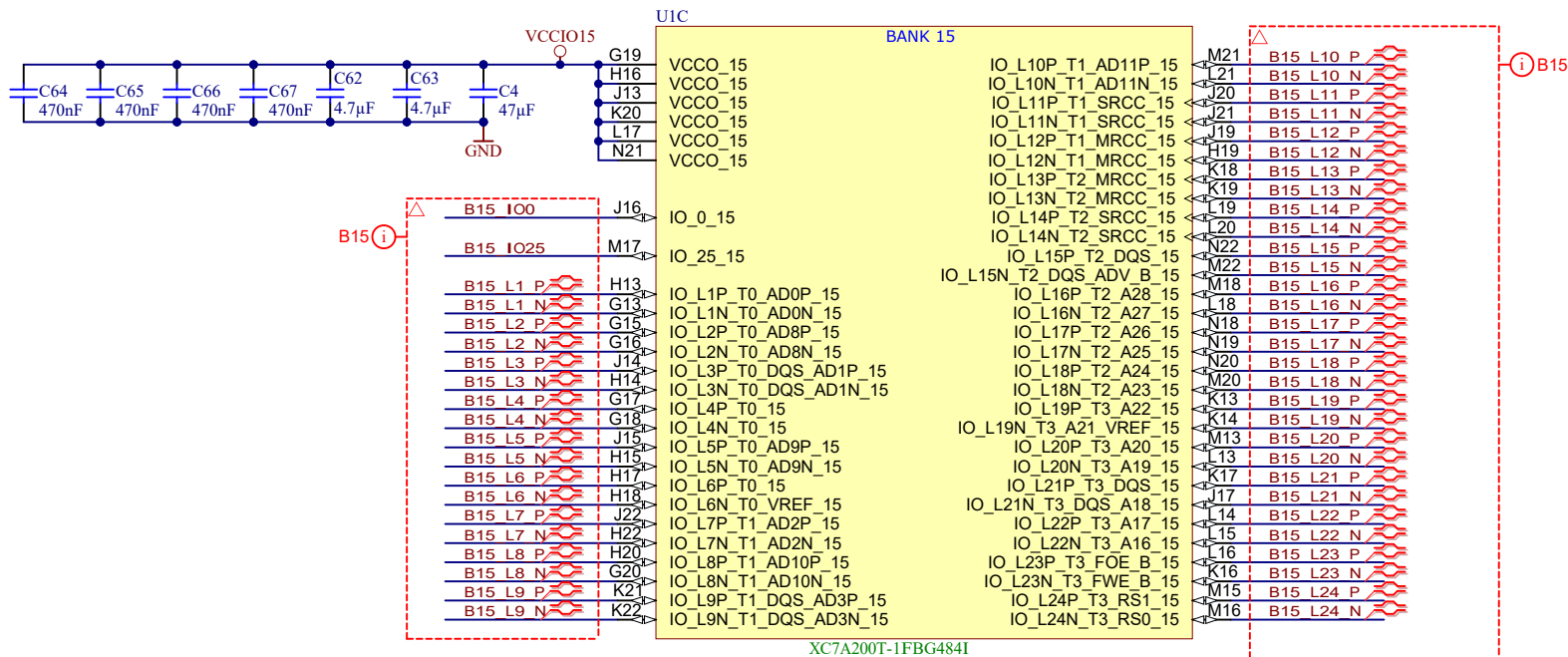
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Date: 29.07.2026	Copyright: Trenz Electronic GmbH	Page 7 of 20
Filename: B13.SchDoc		



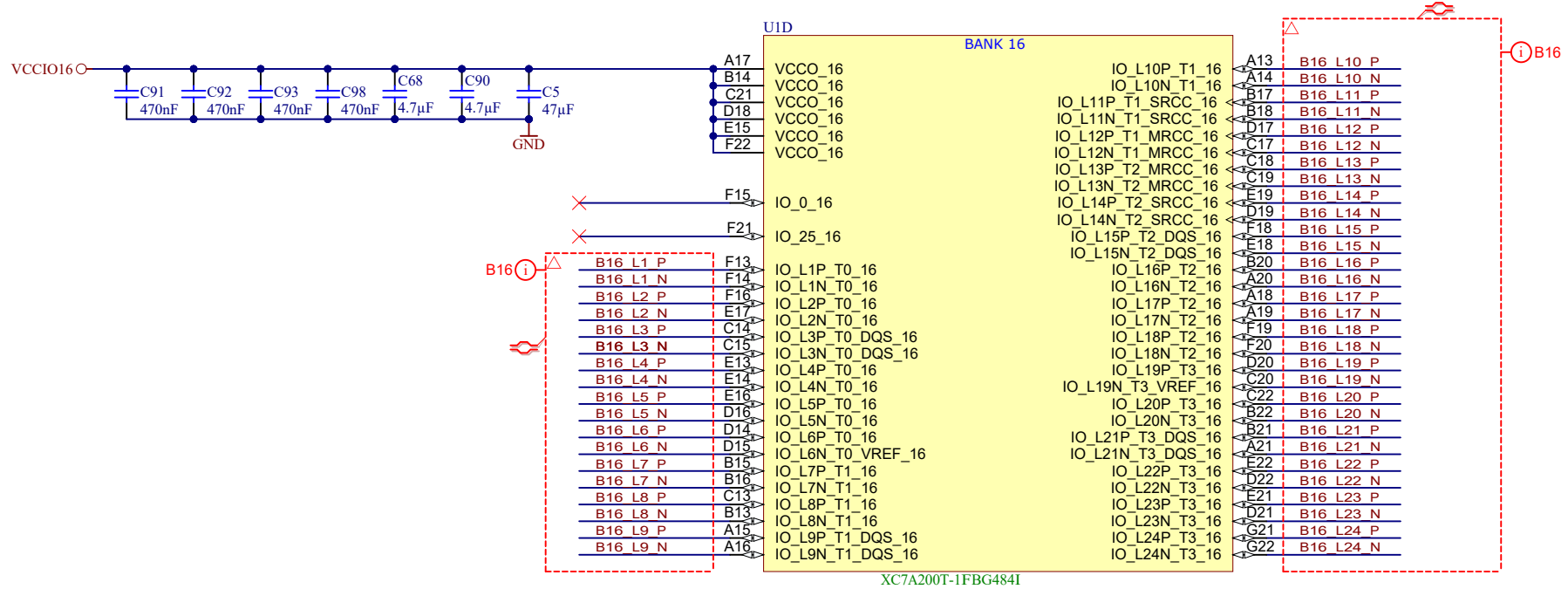
I2C bus addresses		
U1B	FPGA B14	h**
U2	Clock generator	h70
U14	DCDC VCCINT	h61



Title: B14		
A4	Number: TE0712 81136-A	Rev. 03
Date: 30.07.2026	Copyright: Trenz Electronic GmbH	Page 8 of 20
Filename: B14.SchDoc		



	Title: B15		
	A4	Number: TE0712 81136-A	Rev. 03
	Date: 29.07.2026	Copyright: Trenz Electronic GmbH	Page 9 of 20
	Filename: B15.SchDoc		



Title: B16			
A4	Number: TE0712 81I36-A		Rev. 03
	Date: 29.07.2026	Copyright: Trenz Electronic GmbH	Page 10 of 20
Filename: B16.SchDoc			

1

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A

A

B

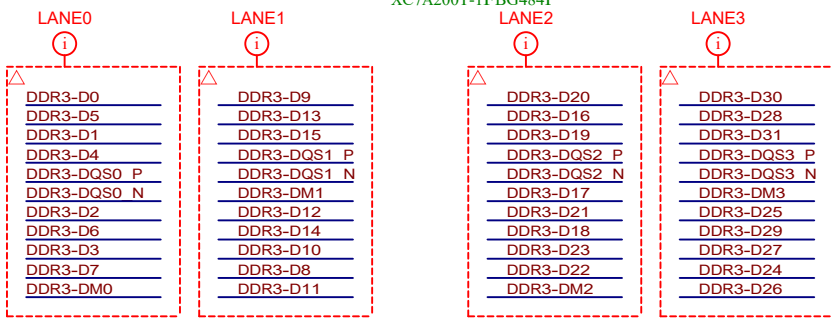
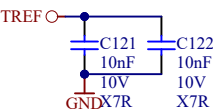
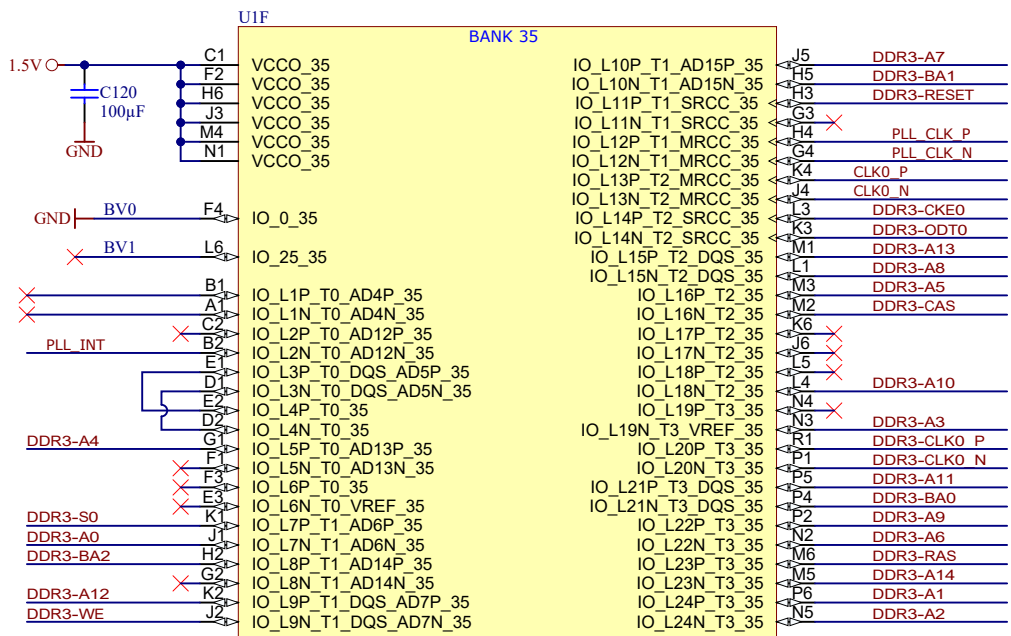
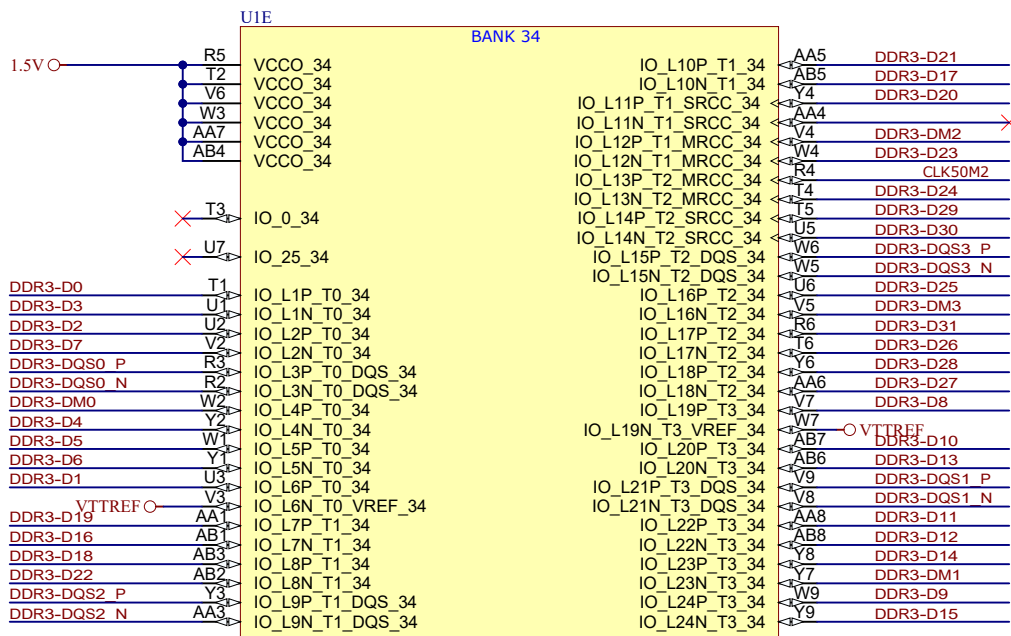
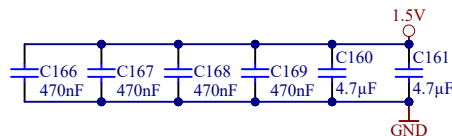
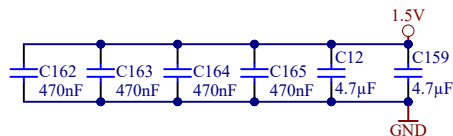
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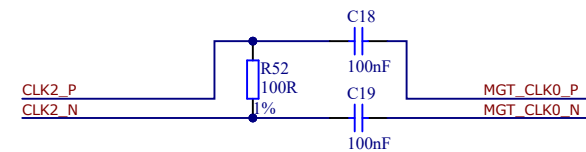
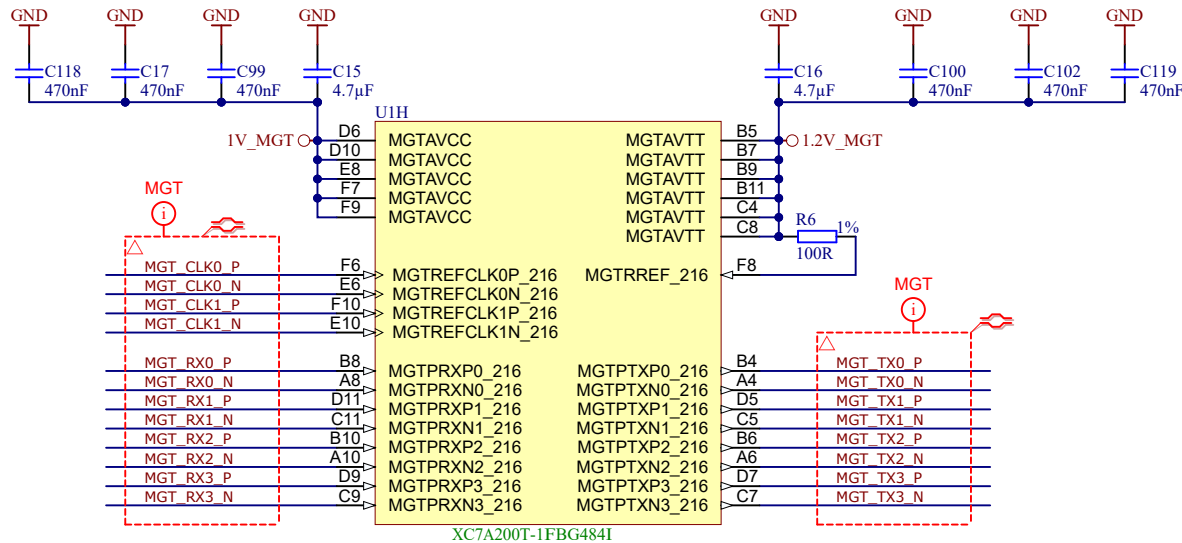
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Date: 30.07.2026	Copyright: Trenz Electronic GmbH	Page 11 of 20
Filename: B34.SchDoc		

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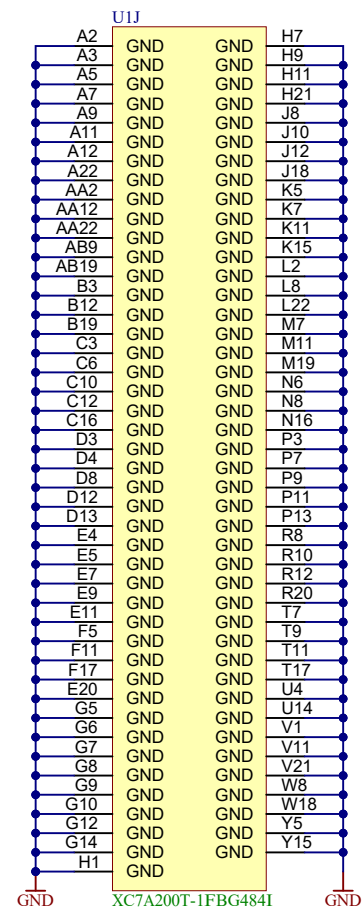
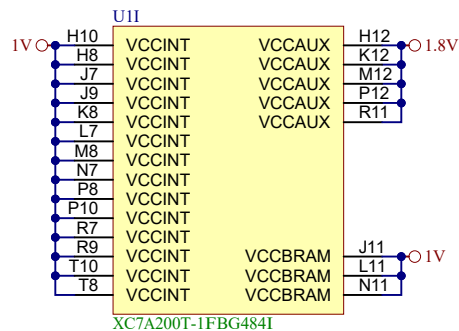
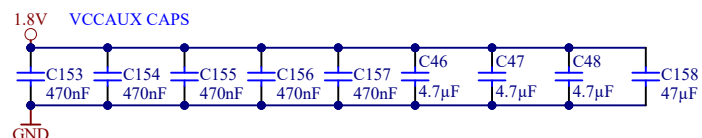
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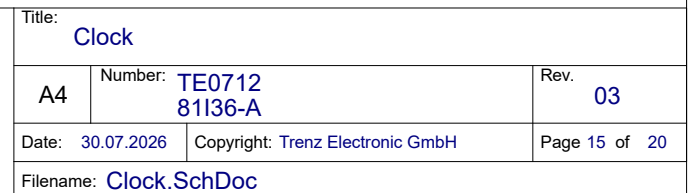
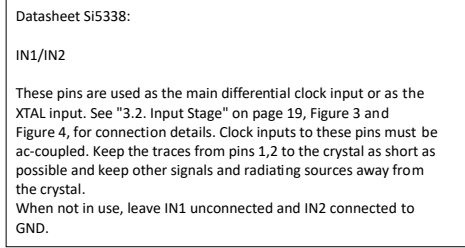
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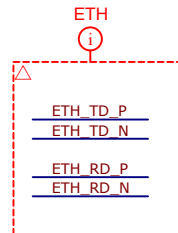
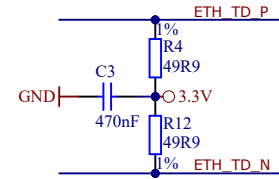
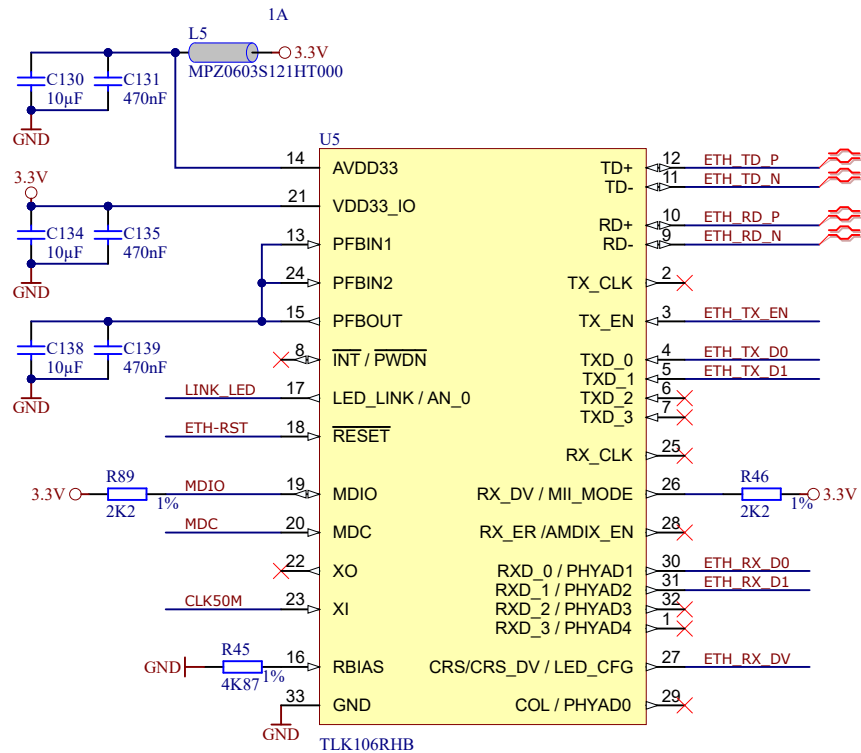
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		A4	Number: TE0712 81136-A
Date: 30.07.2026		Copyright: Trenz Electronic GmbH	
Filename: FPGA-MGT.SchDoc		Page 12 of 20	
		Rev. 03	







Title: ETH		
A4	Number: TE0712 81136-A	Rev. 03
Date: 30.07.2026	Copyright: Trenz Electronic GmbH	Page 17 of 20
Filename: ETHERNET.SchDoc		

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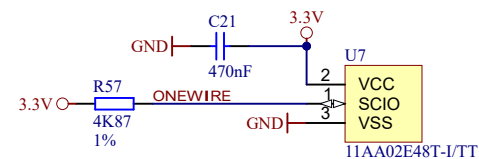
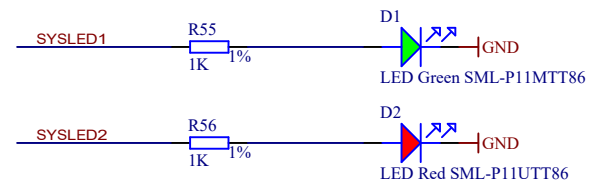
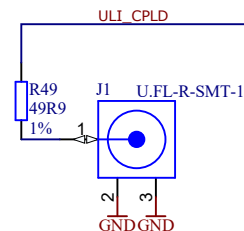
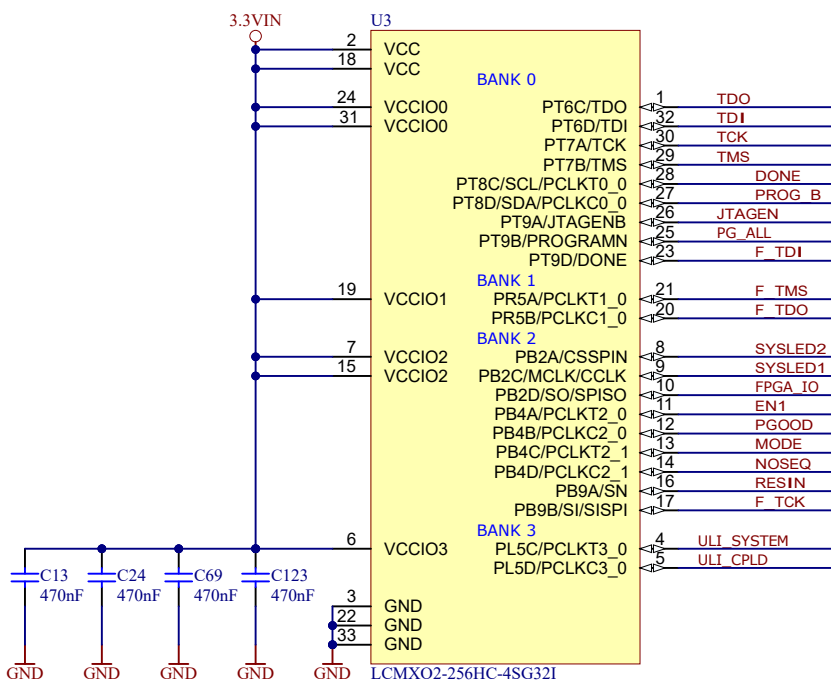
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D



Title: CPLD		
A4	Number: TE0712 81136-A	Rev. 03
Date: 30.07.2026	Copyright: Trenz Electronic GmbH	Page 18 of 20
Filename: CPLD.SchDoc		

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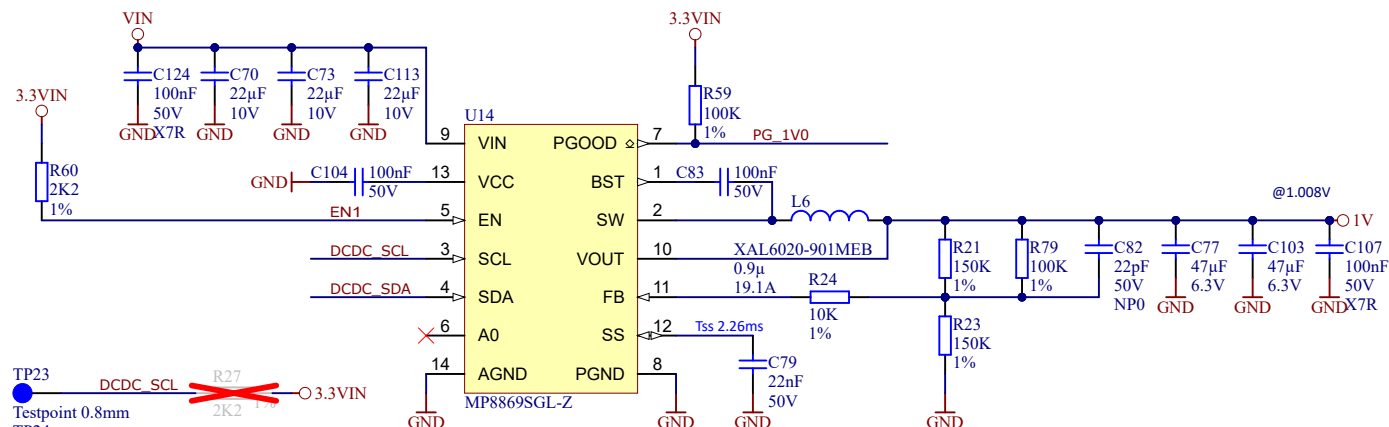
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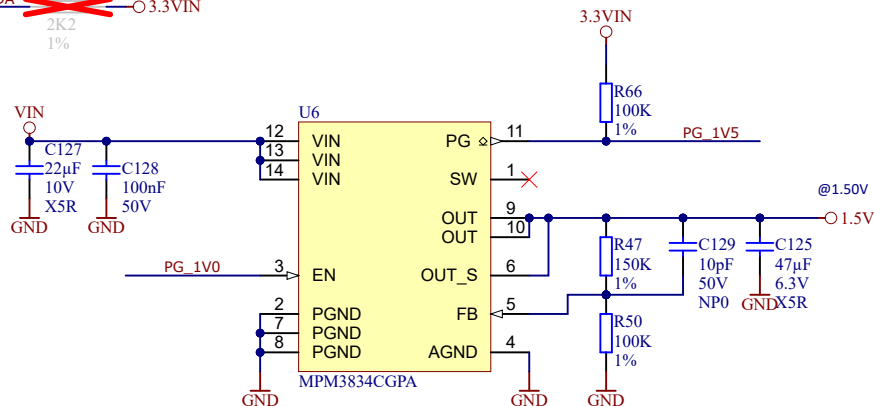
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TP23 Testpoint 0.8mm
TP24 Testpoint 0.8mm

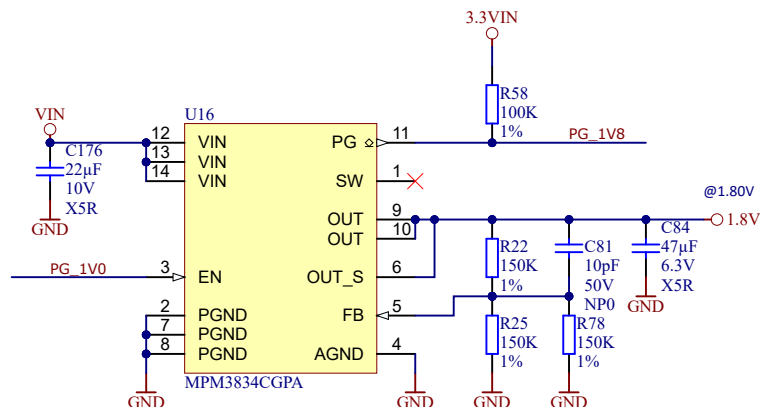


B

B

C

C



D

D



Title: PWR		
A4	Number: TE0712 81136-A	Rev. 03
Date: 30.07.2026	Copyright: Trenz Electronic GmbH	Page 19 of 20
Filename: PWR1.SchDoc		

- VIN TP1 Testpoint 0.8mm
- 3.3VIN TP4 Testpoint 0.8mm
- 3.3V TP7 Testpoint 0.8mm
- 1.8V TP10 Testpoint 0.8mm
- 1.5V TP13 Testpoint 0.8mm
- VTTREF TP3 Testpoint 0.8mm
- 1V TP2 Testpoint 0.8mm
- 1.2V_MGT TP5 Testpoint 0.8mm
- VCCIO13 TP8 Testpoint 0.8mm
- VCCIO15 TP11 Testpoint 0.8mm
- VCCIO16 TP14 Testpoint 0.8mm
- GND TP6 Testpoint 0.8mm
- GND TP9 Testpoint 0.8mm
- GND TP12 Testpoint 0.8mm
- GND TP15 Testpoint 0.8mm
- GND TP16 Testpoint 0.8mm

Current limit 3.28A
Soft start 1.2ms

